Contents

1. Purpose

1.1 This SOP describes the procedure for calibrating laboratory pH meters prior to analytical measurements.

1.2 Calibration ensures measurement accuracy and consistency between laboratory personnel.

2. Scope

2.1 This procedure applies to all benchtop pH meters used within the laboratory.

3. Required Materials

- pH meter
- Calibration buffers (4, 7, and 10)
- Deionized water
- Kimwipes

4. Procedure

4.1. Initial Inspection

4.1 Inspect the electrode for cracks, salt buildup, or air bubbles.

4.2 Verify the electrode storage solution is filled appropriately.

4.2. Calibration

4.3 Rinse the probe using deionized water.

4.4 Immerse the electrode into the pH 7 buffer solution.

4.5 Wait for the reading to stabilize before accepting calibration.

4.6 Repeat the process using pH 4 and pH 10 buffers.

5. Documentation

5.1 Record calibration date, operator initials, and slope values in the laboratory logbook.

6. References

1. Manufacturer operating manual

2. EPA Method 150.1